# Literature Review

Adaptive Multi-Granularity Hierarchical Indexing for

Cost-Efficient Retrieval in Heterogeneous Documents

---

## 1. About Our Project

### Project Title

**Adaptive Multi-Granularity Hierarchical Indexing for Cost-Efficient Retrieval in Heterogeneous Documents**

### What Problem Are We Solving?

Real-world documents are rarely uniform. A single PDF can contain pages of plain text, followed by a page with three large tables, followed by another page with diagrams and images. When AI s ystems process such documents for question answering, they currently treat every page the same way. Every page gets broken into fixed-size chunks and indexed identically, regardless of how si mple or complex that page actually is.

This uniform approach creates two serious problems. First, it wastes computational resources and money because complex pages that need fine-grained indexing are underrepresented, while simpl e pages get indexed at an unnecessarily deep level. Second, retrieval accuracy suffers because the index structure does not reflect the actual information density of the document.

**Our project directly challenges this assumption. We ask a simple but important question: why should every page be indexed the same way when pages are clearly not the same?**

### Our Core Idea

The system works through four steps:

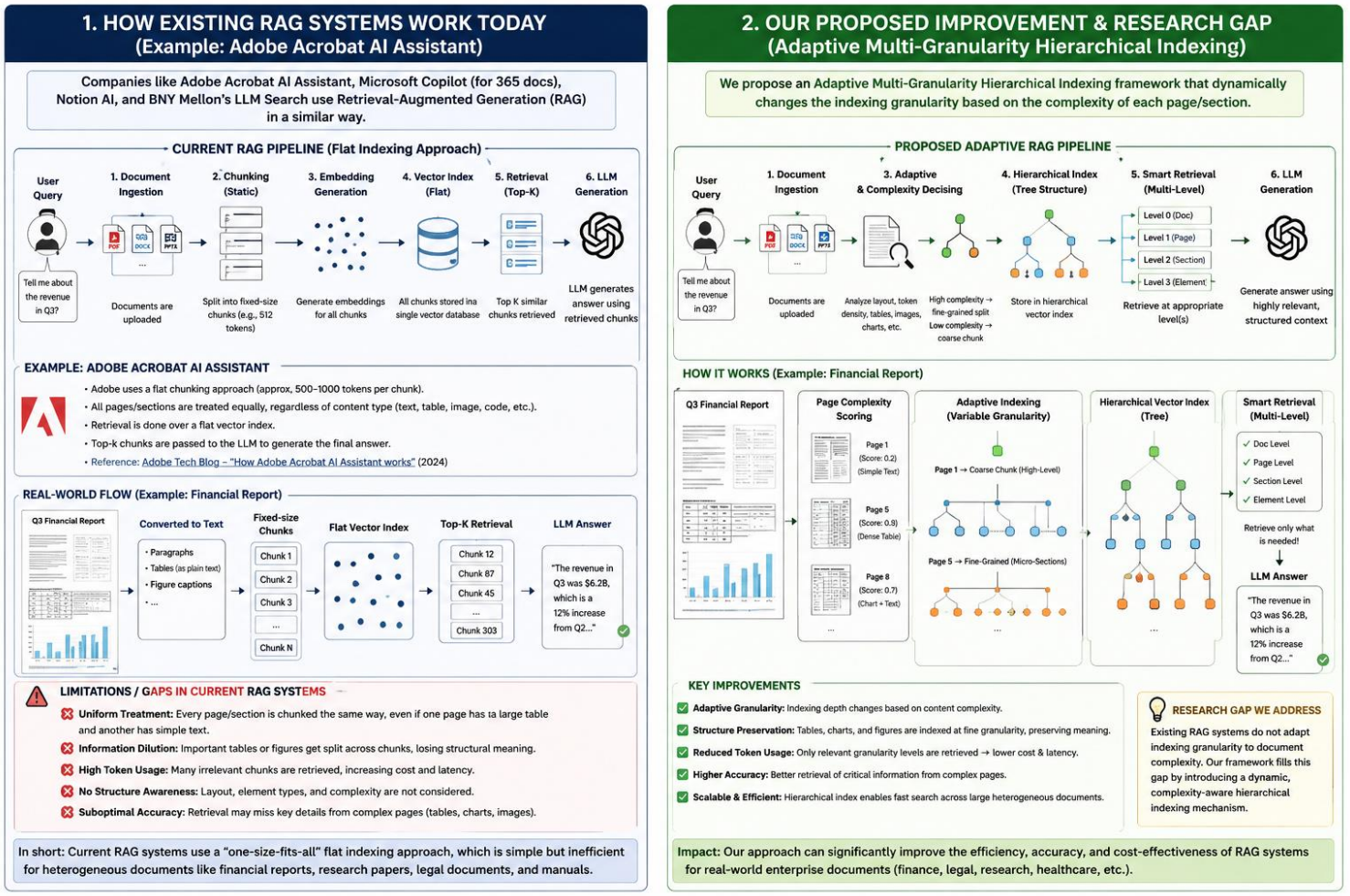
- Each page of the document is individually examined for structural complexity. The system measures token density, the number of tables, the number of images, and layout intricacy.
- Based on this complexity score, an adaptive tree-building algorithm decides how deeply to index that specific page.

**A hierarchical index is built where simple pages receive a lightweight coarse node and complex pages receive a deep subtree with fine-grained child nodes capturing every structural element.**

- During retrieval, the system navigates this hierarchy efficiently, fetching exactly what is needed and reducing token usage and API cost without losing accuracy.

## What Makes This Novel?

**No existing system constructs the index structure itself based on per-page complexity. Prior work either uses fixed hierarchies decided in advance or adapts retrieval routing after indexing is already done. We intervene at index construction time, which is the missing piece in the current literature.**




## 2. Applications of This Project

One of the first things I thought about when designing this system was where it would actually be used in the real world. The answer turned out to be almost everywhere that professionals deal wit h large, mixed-content documents on a daily basis. Here are the most significant application area s.

### Legal and Contract Intelligence

Legal documents are a perfect example of the problem we are solving. A contract might have pag es of standard boilerplate text followed by a page with a complex indemnity clause buried inside a multi-row table. Uniform indexing treats both pages the same. Our system would give the table page deep indexing and the boilerplate page a shallow node, making clause retrieval faster and m ore accurate for legal teams and cutting the cost of running large-scale document discovery.

## Medical and Clinical Records

Patient records are fundamentally heterogeneous. Lab test results appear in structured tables, whi le doctor observations are written in free-form narrative text, and diagnostic images appear along side both. When a clinician or AI assistant queries such a record, it needs to retrieve the right typ e of content from the right level of the index. Our framework naturally handles this by assigning deeper indexing to data-heavy pages and lighter indexing to narrative ones.

## Financial Report Analysis

Annual reports and audit statements are notoriously dense. A single document might run hundred s of pages, with balance sheets, cash flow tables, and management commentary all mixed togethe r. Retrieving a specific financial figure from a dense table requires finer indexing than finding a s ummary paragraph. Our adaptive system allocates resources where they matter and saves cost wh ere they do not.

## Engineering and Technical Documentation

Product manuals, software documentation, and engineering specifications mix wiring diagrams, s pecification tables, code samples, and explanatory prose. Maintenance engineers and AI-powered support systems need to retrieve precise technical details quickly. Adaptive hierarchical indexing ensures that diagram-heavy and table-heavy pages get the depth they need while introductory and overview pages stay lightweight.

## Academic Research Assistance

Research papers and textbooks are another strong use case. A 200-page textbook has sections tha t are simple narrative and sections that are equation-dense or figure-heavy. A student or researche r querying such a document needs the system to understand that the equation on page 47 lives at a different level of representational detail than the introductory paragraph on page 3. Our framewor k handles this distinction automatically.

## Government and Policy Documents

Regulatory notices, policy papers, and government gazettes contain a mix of statutory tables, ann exures, and explanatory text. Accurate and cost-efficient retrieval from such documents supports citizen services, legal compliance platforms, and public policy research, particularly important in environments where every API call carries a cost.

### E-Discovery and Compliance Auditing

In legal investigations or corporate compliance audits, thousands of heterogeneous documents mu st be rapidly searched for specific evidence. Applying our adaptive indexing system to such colle ctions would dramatically reduce processing costs because simple documents and simple pages w ould not consume the same resources as genuinely complex ones. Speed and cost-efficiency both improve.

### Intelligent Tutoring and Education

Educational platforms that answer student questions from textbooks and lecture slides need to ret rieve content at the right level of detail. A question about a formula needs element-level precision while a question about a concept can be answered from a page-level summary. Our hierarchy nat urally supports both without requiring the system designer to hard-code which level to use.

## 3. Related Work

As part of preparing this literature review, I read through seven papers that are closely related to o ur project. Each one taught me something useful, and each one also left a specific gap that our pr oject fills. I have written about each paper below in the order I read them, covering what they do, what I found valuable, and where they fall short for our purpose.

### Paper 1 SeerRAG: Seer-Like Routing for Uncertainty-Guided RAG

---

**Authors:** Rui Li, Shuai Yuan, Qing Ji, Xiaolong Ge

**Publisher:** IEEE

## What the Paper Is About

When I read SeerRAG, the core problem it addresses immediately stood out to me. Most RAG sy stems retrieve information for every single query, whether they need to or not. If a user asks a sim ple factual question that the language model already knows, the system still triggers retrieval, wa stes time, and adds noisy evidence to the context. SeerRAG fixes this by adding an uncertainty es timator that checks whether retrieval is actually necessary before doing it.

Beyond that, SeerRAG also handles documents that mix text and images. Instead of always routi ng every query to the same retrieval path, it has a Modality Discrimination engine that decides w hether a query needs text retrieval, image retrieval, or both. A Cross-modal Synergistic Generatio n module then aligns the evidence from different modalities before generating the final answer. O n the LongDocURL benchmark, the system boosted model accuracy by around 120% and succes sfully filtered out 97% of unnecessary retrieval calls.

## What I Got From This Paper for Our Project

Reading SeerRAG made me think carefully about where adaptation happens in a RAG pipeline. S eerRAG adapts at the very end of the pipeline, at query time, deciding whether to retrieve at all. What it does not do is go back to the beginning and ask whether the index was built well in the fi rst place. It assumes the index already exists in a fixed form.

**SeerRAG proves that selective, intelligent behaviour is better than blanket processing. We take that same philosophy and apply it one step earlier, at index construction time, not query time.**

**The gap it leaves:** SeerRAG does not investigate how documents should be decomposed and stored. The index is static and uniform. This is exactly the problem our project solves.

**Relevance:** *High. Supports the motivation for adaptive, intelligent processing in RAG systems.*

## Paper 2 RAG Based on Dynamic Knowledge Indexing and Semantic Alignment Optimization

---

**Published In:** 2026 International Conference on AI and Social Network Systems (AISNS)

### What the Paper Is About

This paper was probably the most directly relevant one I read. It attacks a problem that is very clo se to ours. Traditional RAG systems use fixed-length text chunks, which break documents at arbi trary boundaries without any understanding of meaning. A paragraph gets cut in half. A sentence that starts one chunk and ends another loses its context. The result is that retrieved content is frag mented and often misaligned with the query.

The paper proposes two main solutions. First, an adaptive chunking algorithm that analyses sema ntic coherence. If two adjacent paragraphs are about the same topic, they stay together. Second, a prompt engineering technique that generates hypothetical question and answer pairs to bring the s emantic space of queries and knowledge closer together. On HotpotQA, the system achieves 37.0 1 Exact Match and 47.53 F1. On SQuAD, it hits 35.12 EM and 45.69 F1, all notably better than b aselines.

### What I Got From This Paper for Our Project

**This paper gave me a really solid foundation for arguing that fixed chunking is harmful. It has the experimental numbers to back it up. When I write our paper's motivation section, I can cite this work as direct evidence that how you decompose a document before indexing it has a measurable impact on retrieval quality.**

However, after reading it carefully, I noticed that their adaptation is based purely on semantic sim ilarity between text blocks. They ask: are these two paragraphs about the same topic? If yes, keep them together. What they do not ask is: what is the structural character of this page? Does it have images? How many tables are on it? How dense is the layout? Those questions are completely ab sent.

**The gap it leaves:** Their system adapts to what the text says but not to what the page looks like. A page with five tables and three images would be treated the same as a plain text page if the text happens to be semantically coherent. Our system catches this.

**Relevance:** *Very High. Direct evidence that adaptive indexing outperforms fixed chunking. Our work extends the principle to structural complexity.*

## Paper 3 MGA-KGF: Multi-Granularity Adaptive Knowledge Graph Fusion for Smart Grid Data

---

### What the Paper Is About

MGA-KGF comes from a completely different domain, power engineering and smart grids. Smar t grids produce enormous amounts of data from multiple sources: real-time sensor readings, trans action logs, equipment maintenance records, and long-term planning data. These data types differ

in timing, structure, and meaning, which makes it very hard to build a single intelligent system on top of them.

MGA-KGF proposes a layered knowledge graph that organises everything into three tiers: fine-grained, medium-grained, and coarse-grained knowledge units. These units are aligned spatially and temporally so that the system can reason across scales. A load-aware storage mechanism dynamically shifts data between hot and cold storage to balance query speed with compression.

## What I Got From This Paper for Our Project

At first glance this paper might seem unrelated to document retrieval, and in terms of domain it i s. But what I found valuable here is the conceptual validation: the idea of representing the same i nformation at different granularity levels, fine, medium, and coarse, is useful and proven to work in heterogeneous data environments.

**Our project does the same thing but for document pages. Simple pages get a coarse representation. Complex pages get fine-grained representation with deep subtree nodes. MGA-KGF demonstrates that this layered philosophy scales and works in practice, even if the domain is different.**

**The gap it leaves:** It is a data fusion paper, not a document retrieval paper. None of its techniques transfer directly to our setting. But it lends cross-domain credibility to our approach.

**Relevance:** *Moderate. Validates multi-granularity representation as a general principle across heterogeneous data domains.*

## Paper 4 Multigranularity Information Fusion for Multimodal Retrieval in Prefabricated Construction

---

**Publisher:** IEEE Industrial Informatics

### What the Paper Is About

This paper was a genuinely interesting read because it operates in exactly the kind of domain that motivates our project: documents that are difficult because they mix text, tables, and images in w ays that general-purpose models cannot handle well. The setting is prefabricated construction, do cuments that contain technical specifications, multilingual content, and rich visual elements.

The proposed framework trains embedding models using three objectives at the same time: a page-level contrastive loss that preserves broad contextual meaning, an element-level contrastive loss that captures fine-grained detail inside tables and images, and an answer-guided auxiliary loss that pushes the embeddings toward downstream query relevance. The result is a system that is better at both broad and specific retrieval.

### What I Got From This Paper for Our Project

Reading this paper, I felt it was the closest existing work to what we are proposing, and understan ding exactly where it differs from us helped me sharpen our novelty claim significantly.

The paper explicitly recognises that page-level and element-level representations complement ea ch other. That is the same intuition behind our hierarchical index. Where they differ from us is th at they fix the granularity levels at exactly two: page and element. Every document, regardless of how simple or complex any individual page is, gets processed at both levels.

**Both systems recognise that a single granularity is insufficient for heterogeneous documents. Both use multiple levels of representation. This is the core insight we share.**

**The gap it leaves:** Their two-level structure is rigid. A page with one small table gets the same element-level indexing as a page with five tables and multiple diagrams. Our system assigns different depths to these two pages because their complexity scores would differ. This selective per-page depth assignment is our contribution beyond this paper.

**Relevance:** *Very High. Strongest conceptual neighbour to our work. Our project extends fixed two-level granularity to adaptive per-page depth.*

## Paper 5 LLM-Centric RAG with Multi-Granular Indexing and Confidence Constraints

---

### What the Paper Is About

This paper proposes a RAG system that stores knowledge in a layered memory structure spanning multiple granularities, from fine-grained sentence level all the way up to the full document level. Rather than retrieving from a single flat index, the system can access whichever memory layer best matches the query. During generation, a confidence estimation module checks the retrieved evidence and filters out low-confidence retrieval paths before they reach the language model. The result is a reduction in hallucinations and improvements in QA accuracy, Recall@5, NDCG@5, and factual consistency.

### What I Got From This Paper for Our Project

**What I appreciated most about this paper is what it proves empirically: multi-granularity indexing is better than single-granularity indexing. The numbers back this up clearly. This is one of the strongest pieces of evidence I can cite in our paper to justify why building a hierarchical index is worth the engineering effort.**

But when I looked at how the hierarchy was built, I noticed something important: it is fixed. The authors define the levels, sentence, paragraph, and document, and apply them uniformly to all co

ntent. They do not ask whether a given page actually needs all those levels or whether the levels s hould be determined by the page's structural properties.

**The gap it leaves:** They assume the hierarchy already exists in a fixed form. Our project answers the question they never asked: how should the hierarchy be built differently for different pages?

**Relevance:** *High. Proves multi-granularity indexing beats single-level. Our work contributes the adaptive construction of that hierarchy.*

## Paper 6 AMR-CSI: Adaptive Multimodal RAG for Cold Start Indexing

---

**Authors:** Siva Prasad et al.

**Conference:** MMGR 2025 (ACM)

### What the Paper Is About

AMR-CSI addresses something called the cold-start indexing problem: when a brand-new docum ent collection arrives and needs to be indexed from scratch, how should the system decide what k ind of index to build? Should it index the collection as text, or as visual pages? The paper answer s this by calculating a Visual Dominance Score for each collection based on layout analysis. If th e score is high, the system builds a visual index. If the score is low, it builds a text-based index. T his automated routing decision outperforms applying a single uniform indexing strategy to everyt hing.

### What I Got From This Paper for Our Project

I found this paper genuinely exciting to read because it is the closest work in the literature to wha t we are proposing. Both papers start from the same realisation: you should analyse the document

first, make an intelligent decision, and then build the index accordingly. That is a fundamentally d ifferent philosophy from the standard approach of indexing everything the same way.

**This paper proves the principle that pre-indexing document analysis and adaptive indexing decision-making is valid and effective. That is exactly the hypothesis our project builds upon.**

The difference between AMR-CSI and our project comes down to granularity and depth of decisi on:

- AMR-CSI makes one decision per collection: which modality index to use. The decision is binary, text or visual.
- **Our project makes one decision per page: how many levels of hierarchical depth to assign. The decision is continuous, from a single coarse node up to a deep multi-level subtree.**

**The gap it leaves:** A collection might be classified as text-dominant overall while containing individual pages that are image-heavy and desperately need deeper indexing. AMR-CSI would miss this. We would not.

**Relevance:** *Very High (closest existing work). Proves adaptive indexing beats uniform strategies. Our work extends this to page-level depth decisions.*

## Paper 7 Multi-Granularity Retrieval Framework for Visually Rich Documents

---

### What the Paper Is About

Visually rich documents, those containing text, tables, charts, images, and complex layouts, cannot be effectively retrieved using a single chunk representation. This work proposes a framework that indexes information at multiple scales: entity level, sentence level, paragraph level, page level, and document level. During retrieval, the system accesses the most appropriate level based on the query. The multi-scale architecture improves both precision and recall for heterogeneous document types.

## What I Got From This Paper for Our Project

This paper supports our central argument that not all information should be stored or retrieved at t he same granularity. It demonstrates that multi-scale retrieval for visually rich documents improv es outcomes and provides good empirical backing that is directly applicable to the kind of hetero geneous documents our project targets.

**Both systems recognise that multi-granularity access is needed for heterogeneous documents. We share the same core philosophy and this paper supports our argument.**

Where I see its limitation is in the cost and efficiency of indexing everything at all five levels for every page. Consider a document where 60% of the pages are straightforward narrative text. Inde xing each of those pages at five granularity levels is expensive and mostly unnecessary. The entit y-level and sentence-level indexes for a page that contains two simple paragraphs will not add me aningful retrieval value.

**The gap it leaves:** Indexing every page at every level regardless of complexity wastes resources. Our system would assign those simple narrative pages a single coarse node and reserve deep multi-level indexing for pages that genuinely need it. This selective assignment is where our efficiency gains come from.

**Relevance:** *High. Multi-scale retrieval for visually rich documents. Our work adds complexity-driven selective depth to avoid over-indexing.*

## 4. Research Gap

After reading through all seven papers, the pattern that emerged was actually quite satisfying fro m a research perspective. The community has made real progress on pieces of this problem, adap tive retrieval routing, adaptive chunking, multi-granularity representation, and adaptive modality selection. But no one has put these ideas together in the specific way we propose.

Here is what each paper adapted and what it did not:

- SeerRAG adapted retrieval routing at query time but left the index static.
- The dynamic knowledge indexing paper adapted chunk boundaries based on text semantics but ignored visual and structural page features.
- MGA-KGF used fine, medium, and coarse granularity for smart grid data but did not work in the document retrieval domain at all.
- The multigranularity construction retrieval paper used page and element levels but fixed both levels and applied them identically to every document.
- The LLM-centric RAG paper exploited a pre-existing multi-level hierarchy effectively but never asked how to build it adaptively.
- AMR-CSI made an adaptive indexing decision at the collection level to select modality, not per page, and not about index depth.
- The visually rich document retrieval framework indexed all pages at all five granularity levels, which is accurate but computationally expensive.

**The specific gap is this: no existing work dynamically constructs the depth of the document index on a per-page basis based on structural complexity. This is what we contribute. We measure the actual complexity of each page, how many tables it has, how many images, how dense the text, how intricate the layout, and we build the hierarchical index to match.**

**Complex pages get deep. Simple pages stay shallow. The result is a system that is both more accurate on complex content and more cost-efficient on simple content.**

## 5. Summary of Reviewed Papers

---

The following is a quick reference of all seven papers, their relevance to our project, and the spec ific gap each one leaves.

**Paper 1** SeerRAG (IEEE)

- What they do: Adaptive retrieval routing using uncertainty estimation at query time.
- What we get from it: Evidence that selective processing beats indiscriminate retrieval.
- What it does not do: The index remains fixed and uniform. No adaptation at construction time.

**Paper 2** Dynamic Knowledge Indexing (AISNS 2026)

- What they do: Adaptive chunking based on semantic coherence between text blocks.
- What we get from it: Proof that fixed-length chunking is harmful with experimental numbers to cite.
- What it does not do: Does not account for visual features, table density, or structural complexity.

**Paper 3** MGA-KGF

- What they do: Fine, medium, and coarse granularity knowledge graph for smart grid data fusion.
- What we get from it: Cross-domain validation that multi-granularity representation works.
- What it does not do: Entirely different domain. No direct transfer to document retrieval.

**Paper 4** Multigranularity Fusion for Prefabricated Construction (IEEE)

- What they do: Page-level and element-level retrieval for mixed-content construction documents.
- What we get from it: Strong conceptual neighbour showing two-level granularity improves retrieval.
- What it does not do: Two levels are fixed and applied to all pages equally. No adaptive depth.

**Paper 5** LLM-Centric RAG with Confidence Constraints

- What they do: Multi-granularity hierarchical memory with confidence filtering during generation.
- What we get from it: Empirical proof that multi-level indexing outperforms single-level indexing.
- What it does not do: The hierarchy is assumed to exist. Construction is not addressed.

**Paper 6** AMR-CSI (ACM MMGR 2025)

- What they do: Adaptive modality selection at collection level during cold start indexing.
- What we get from it: The strongest evidence that pre-indexing document analysis improves outcomes.
- What it does not do: One decision per collection. Does not decide per-page index depth.

**Paper 7** Multi-Granularity Retrieval for Visually Rich Documents

- What they do: Five-level multi-scale indexing and retrieval for heterogeneous document content.
- What we get from it: Supports the argument that multi-scale access improves retrieval quality.
- What it does not do: All pages indexed at all five levels regardless of complexity. Costly and indiscriminate.